

► 1853

FIRST MANNED GLIDER

English engineer Sir George Cayley is the first person to understand the forces acting upon an aircraft wing. In 1853 he transports his coachman across a small valley in what he calls a "governable parachute" – the first man-carrying glider.

AS A RESULT OF HIS RESEARCH, GEORGE CAYLEY IS OFTEN CALLED THE "FATHER OF FLIGHT"

► 1896

OTTO LILIENTHAL

After making over 2,000 glides in weight-shift controlled gliders, German pioneer Otto Lilienthal dies in hospital after his glider stalls and he crashes from a height of 15 m (50 ft). His scientific data on flight would inspire many others.



OTTO LILIENTHAL'S HANG-GLIDER

► 1900



1928 ZEPPELIN

FIRST AIRSHIP

LZ1, the first rigid airship (designed by Ferdinand, Graf von Zeppelin), makes its initial flight from a floating hangar on Lake Constance near Friedrichshafen, Germany. Carrying five people, it stays airborne for 17 minutes.

► 1903

FIRST POWERED FLIGHT

The first controlled, powered aeroplane flight is achieved by American inventors the Wright brothers. It lasts just 12 seconds and covers 36.5 m (120 ft).



WRIGHT FLYER

► 1909

FIRST CHANNEL CROSSING

Flying his Type XI monoplane, Frenchman Louis Blériot crosses the English Channel for the first time in a heavier-than-air aircraft. He crash-lands in a field above the cliffs of Dover on the English coast.

◀ 1930

AMY JOHNSON

The first woman to fly solo from England to Australia makes the journey in a Gipsy Moth named "Jason". With only experience of flying from London to Hull in England, Amy Johnson makes her epic 18,000 km (11,000 miles) trip in a small, low-powered biplane more suited to club flying.



GIPSY MOTH

◀ 1927

FIRST TRANSATLANTIC SOLO FLIGHT

American pilot Charles Lindbergh takes 33.5 hours to complete the first solo, nonstop, transatlantic flight, travelling from New York to Paris. Flying in a single-engine aircraft, he encounters fog and icy conditions, though his biggest challenge is staying awake for the entire journey.

◀ 1919

FIRST TRANSATLANTIC FLIGHTS

In May an NC-4 commanded by Albert C Read crosses the Atlantic in several stages from Long Island, USA, to Portugal. In June John Alcock and Arthur Brown fly nonstop from Newfoundland to Ireland.



CURTISS NC-4

◀ 1917

EUGENE JACQUES BULLARD



FIRST BLACK COMBAT PILOT

Georgia-born Eugene Jacques Bullard – who was denied entry into the US Army Air Corps because of his race – serves throughout World War I in the French Flying Corps. He is awarded the Legion of Honour.

◀ 1910

FIRST TAKEOFF FROM A SHIP

In November American flight pioneer Eugene Burton Ely successfully takes off from the deck of a ship. Two months later, he makes the first successful landing aboard a ship.

FOR PROTECTION, ELY WORE A PADDED AMERICAN FOOTBALL HELMET AND A LIFE JACKET MADE FROM PARTS OF BICYCLE TYRES

► 1988

HEAVIEST AIRCRAFT

Designed to transport the Soviet Union's 250-tonne Buran space shuttle, the six-engined Antonov An-225 takes the record for the world's biggest and heaviest aircraft. It also has the largest wingspan of any aircraft in operational service.

THE AN-225 HOLDS THE RECORD FOR AIRLIFTING THE HEAVIEST CARGO



► 1991



F-117 NIGHTHAWK

FIRST STEALTH FIGHTER

The American Lockheed F-117 Nighthawk sees its first active service during Operation Just Cause in Panama.

► 2005



AIRBUS A380

LARGEST MASS-PRODUCED AIRCRAFT

The double-deck, four-engine Airbus A380 is launched. This airliner can transport 853 passengers and can carry enough fuel to fly nonstop from Sydney, Australia, to Dallas, USA, – a distance of 13,804 km (8,577 miles).

► 2015

SOLAR IMPULSE 2 SETS OFF

Using only energy from the Sun, Solar Impulse 2 leaves on an epic round-the-world flight. Its wingspan is larger than a Boeing 747's, though the aircraft weighs no more than a small van.



ANTONOV AN-225